Application: the heat equation

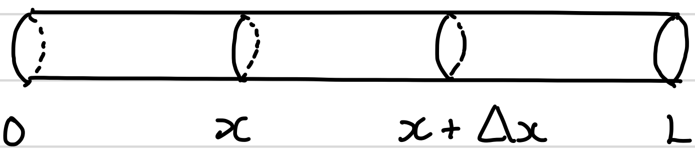
Derivation in 1D

- Suppose that we have a rod of length L, and assume the following:

(1) The rod is made of a single homogeneous conducting material.

(2) Heat flows only in the x-direction (i.e. the rod is laterally insulated).

(3) The temperature is constant at all points of a cross-section (i.e. the rod is thin).



- Now,

Net rate of change of heat inside $[x, x+\Delta x]$

= Net flux of heat across the boundaries (conservation of energy)

- We also have that

Total heat inside $[x, x+\Delta x]$

$$= \int_x^{x+\Delta x} c\rho A u(s,t)\,ds$$

where c is the thermal capacity of the rod (measures its ability to store heat), ρ is its density, A is its cross-sectional area, and u is the temperature distribution.

- We can then write the conservation of energy equation as

$$\overbrace{\frac{d}{dt}\int_x^{x+\Delta x} c\rho A\, u(s,t)\,ds = c\rho A\int_x^{x+\Delta x} u_t(s,t)\,ds}^{\text{Net rate of change of heat}}$$

$$= \underbrace{kA\left[u_x(x+\Delta x, t) - u_x(x,t)\right]}_{\text{Net flux of heat across the boundaries}}$$

where k is the thermal conductivity of the rod (measures its ability to conduct heat) and the flux of heat is proportional to the temperature gradient $u_x = du/dx$.

<u>Recall</u>: <u>the mean value theorem</u>

Suppose that F is continuous on $[a, b]$ and differentiable on (a, b). Then, there exists at least one number ξ in (a, b) such that

$$F'(\xi) = \frac{F(b) - F(a)}{b - a}.$$

or, if $F' = f$, we have $F(b) - F(a) = \int_a^b f(x)\,dx$

and $\int_a^b f(x)\,dx = f(\xi)(b - a)$

- Applying this result,

$$c\rho A\, u_t(\xi, t)\,\Delta x = kA\left[u_x(x+\Delta x, t) - u_x(x, t)\right]$$

$$u_t(\xi, t) = \frac{k}{c\rho}\left\{\frac{u_x(x+\Delta x, t) - u_x(x, t)}{\Delta x}\right\}$$

- Letting $\Delta x \to 0$, we have that

$$\boxed{u_t(x, t) = \kappa\, u_{xx}(x, t)}$$

where $\kappa = k/(c\rho)$.

- This is the <u>heat equation</u>.

<u>Notes</u>:- we can make a similar argument using the divergence theorem to derive the heat equation in higher dimensions:

$$\boxed{\frac{\partial u}{\partial t} = \kappa \nabla^2 u}$$

- The <u>diffusion equation</u>,

$$\boxed{\frac{\partial u}{\partial t} = D \nabla^2 u}$$

where u is the concentration of (e.g.) a gas and D is the diffusion constant, can be derived similarly, using mass conservation instead of energy conservation.

The Laplace equation

- When the temperature has reached a steady state, $\partial u / \partial t = 0$, and the heat equation becomes the Laplace equation:

$$\boxed{\nabla^2 u = 0}$$

<u>Example</u> Solve the 2D Laplace equation using separation of variables.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{becomes}$$

$$X'' Y + X Y'' = 0 \quad \text{or} \quad \frac{X''}{X} = -\frac{Y''}{Y} = \lambda^2,$$

where λ^2 is the separation constant. We then find

$$X'' = \lambda^2 X \quad \text{and} \quad Y'' = -\lambda^2 Y$$

- We first choose λ^2 to be > 0, and try a solution for X of the form $X = e^{mx}$.
- This gives
$$m^2 e^{mx} = \lambda^2 e^{mx},$$
so the auxiliary equation is
$$m^2 - \lambda^2 = 0$$
and has solutions $m = \pm \lambda$.

- We then have
$$X(x) = a e^{\lambda x} + b e^{-\lambda x}$$
or, equivalently,

$$X(x) = A \cosh \lambda x + B \sinh \lambda x$$

where a, b, A and B are all constants.

- Trying a solution for Y of the form $Y = e^{ny}$, we find that the auxiliary equation is $n^2 + \lambda^2 = 0$, so that $n = \pm i\lambda$
- Then, $Y(y) = c e^{i\lambda y} + d e^{-i\lambda y}$ or

$$Y(y) = C \cos \lambda y + D \sin \lambda y.$$

where c, d, C and D are constants.

- A solution of the PDE is then

$$u(x, y) = (A \cosh \lambda x + B \sinh \lambda x)(C \cos \lambda y + D \sin \lambda y)$$

- If we had written the separation constant as $-\lambda^2 < 0$, the roles of x and y would have switched.
- The choice of sign for the separation constant is fixed by the boundary conditions.

<u>Example</u> Solve $\dfrac{\partial u}{\partial t} = 4 \dfrac{\partial^2 u}{\partial x^2}$

with boundary condition (BC) $u(0, t) = u(\pi, t) = 0$

and initial condition (IC) $u(x, 0) = 2 \sin 3x - 4 \sin 5x$

This is a heat equation with $\kappa = 4$ (or a diffusion equation with $D = 4$).

- Let $u(x, t) = X(x) T(t)$. Then,

$XT' = 4 X'' T$ and $\dfrac{X''}{X} = \dfrac{T'}{4T}$

- Both sides must be equal to a separation constant, which we call $-\lambda^2$.
- Then,

$$X'' + \lambda^2 X = 0 \quad \text{and} \quad T' + 4\lambda^2 T = 0$$

so that $X = a \cos \lambda x + b \sin \lambda x$

and $T' = -4\lambda^2 T \Rightarrow \dfrac{dT}{dt} = -4\lambda^2 T \Rightarrow \dfrac{dT}{T} = -4\lambda^2 dt$

$\Rightarrow \ln T = -4\lambda^2 t + C \qquad \Rightarrow T = e^C e^{-4\lambda^2 t}$

so

$$T = ce^{-4\lambda^2 t} \quad \text{with} \quad c = e^C$$

- Choosing $+\lambda^2$ as the separation constant would have given hyperbolic solutions for X and a growing exponential for T.
- The solution would have been unbounded and would not have satisfied the boundary conditions.
- Then, a solution is

$$u(x, t) = XT = ce^{-4\lambda^2 t}(a \cos \lambda x + b \sin \lambda x)$$

$$= e^{-4\lambda^2 t}(A \cos \lambda x + B \sin \lambda x)$$

where $A = ac$ and $B = bc$.

- Since $u(0, t) = 0$, $A = 0$, and $u = B \exp(-4\lambda^2 t) \sin \lambda x$.
- Since $u(\pi, t) = 0$, we must have that $\sin(\lambda\pi) = 0 \Rightarrow \lambda = n = 0, \pm 1, \pm 2, \ldots$
- Then,

$$u(x, t) = B e^{-4n^2 t} \sin(nx)$$

is a solution and, by the principle of superposition,

$$u(x, t) = B_1 e^{-4n_1^2 t} \sin(n_1 x) + B_2 e^{-4n_2^2 t} \sin(n_2 x)$$

is also a solution.
- From the initial condition,

$$u(x, 0) = B_1 \sin(n_1 x) + B_2 \sin(n_2 x)$$

$$= 2\sin 3x - 4\sin 5x$$

$\Rightarrow$ $B_1 = 2$, $B_2 = -4$, $n_1 = 3$ and $n_2 = 5$, so the required solution is

$$u(x,t) = 2e^{-36t}\sin(3x) - 4e^{-100t}\sin(5x)$$